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DETAIL SPECIFICATION

MICROPHONES, INTERCOMMUNICATION AND ANNOUNCING SYSTEMS, NAVAL SHIPBOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers microphones designed for use in intercommunication and announcing systems aboard naval ships. The microphones are noise cancelling types suitable for use in noisy areas.

1.2 Classification. The types of microphones are designated in the following form, as specified (see 6.2).

Type	Description
M124/SIC	Handheld, equipped with 18-inch length of LSMCOS-5 cable and six-pin connector (replacement for type IC/MPM-3).
M125/SIC	Handheld, equipped with 120-inch coil cord and six-pin connector (replacement for types IC/MPM-6 and M-114/U).
M132/SIC	Handheld, equipped with 120-inch coil cord and six-pin connector (replacement for type IC/MPM-4).
M131/WIC	Handheld, equipped with 120-inch coil cord and ten-pin connector (replacement for type M-106/U).
M136/SIC	Handheld, equipped with 24-inch flexible arm consisting of two double-interlocking stainless-steel hoses and brass enclosure (replacement for type IC/MFM-2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5830

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FEDERAL SPECIFICATIONS

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-10544	-	Connectors, Plug and Receptacle (Electrical, Audio, Waterproof, Ten Contact, Polarized)
MIL-I-631	-	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-24558/18	-	Terminal Boxes, Connection, Submersible (15-Foot), 600 Volts, 4-Terminal, 8-Wire, Symbols No. 400.1 and No. 400.4, 4-Terminal, 12-Wire, Symbols No. 400.2 and No. 400.3
MIL-DTL-24643/4	-	Cable, Electrical, -20 °C to +90 °C, 600 Volts, Type LSMCOS

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-202-101	-	Method 101, Salt Atmosphere (Corrosion)
MIL-STD-202-301	-	Method 301, Dielectric Withstanding Voltage
MIL-STD-889	-	Galvanic Compatibility of Electrically Conductive Materials
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

S6502-74117	-	Plug and Receptacle for Announcing Equipment

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVSEA PUBLICATIONS

T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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U.S. GENERAL SERVICES ADMINISTRATION

Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/AMERICAN STANDARD ASSOCIATION (ASA)

ANSI/ASA S1.15/Part 1 - Measurement Microphones – Part 1: Specifications for Laboratory Standard Microphones

(Copies of this document are available online at http://webstore.ansi.org/.)

ASTM INTERNATIONAL

ASTM B139/B139M - Standard Specification for Phosphor Bronze Rod, Bar, and Shapes

ASTM B209/B209M - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

(Copies of these documents are available online at www.astm.org.)

INSTITUTE OF PRINTED CIRCUITS (IPC)

IPC J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies

IPC J-STD-004 - Requirements for Soldering Fluxes

IPC J-STD-006 - Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications

(Copies of these documents are available online at www.ipc.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Description. Microphones shall be handheld types. Each microphone shall consist of either a dynamic or magnetic element and a press-to-talk switch housed in a plastic case with a microphone cable and connector attached as specified herein. Unless otherwise specified (see 6.2), a microphone holder assembly as specified in 3.11 shall be furnished with each microphone.

3.3 Construction and material.

3.3.1 Metal parts. All metal parts, other than current-carrying parts, shall be made of corrosion-resistant material or shall be suitably protected to resist corrosion.

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3.3.1.1 Aluminum. Aluminum used for construction shall conform to alloy 2024, temper T4 of ASTM B209/B209M; alloy 5052, temper T4 of ASTM B209/B209M; or other materials specifically approved as specified (see 6.2).

3.3.1.2 Dissimilar metals. Selection of metals for use in electronic equipment should be made in accordance with the requirements of MIL-STD-889. Where the requirements herein preclude the insulation of incompatible metal combinations, as identified in MIL-STD-889, from one another, the metal combination shall be isolated by plating or coating, and the contact surface shall be insulated from exterior environments.

3.3.1.3 Fasteners. Fastener hardware shall be made of material that is anti-corrosive or austenitic corrosion-resistant steel and to commercial specifications. Finished plating is not required on stainless steel.

3.3.2 Special tools. Microphones shall be constructed so that special tools will not be required for disassembly, repair, or reassembly. Special tools are defined as tools not listed in the GSA Global Supply Catalog.

3.3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.4 Prohibited materials. The microphones shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.3.5 Solder connections. The soldered electrical connections shall be manufactured and inspected in accordance with class 2 or 3 guidelines of IPC J-STD-001. Solder and solder flux shall be in accordance with IPC J-STD-004 and IPC J-STD-006, respectively.

3.4 Form factor. The microphone case shall fit and lock into the MT-3400/SIC microphone holder assembly in 3.11 (see figure 1). The microphone case shall fit within the IC/MSB-2 microphone station with its door closed.

3.5 Plastic material. The plastic material used in making the microphone case shall be able to withstand all of the tests specified herein without evidence of physical damage. Plastic material shall be black.

3.6 Replaceable parts. The microphone design shall allow easy replacement of the press-to-talk switch and the microphone cable.

3.7 Cables and connector plugs. Microphones shall be equipped with cables and connectors as specified in 3.7.1 through 3.7.3. The cable shall be secured by a clamp in the microphone case to prevent strain on the microphone electrical terminals.

3.7.1 Microphone type M124/SIC. The type M124/SIC microphone shall be equipped with an 18-inch length of LSMCOS-5 cable in accordance with MIL-DTL-24643/4 and a microphone connector plug in accordance with S6502-74117.

3.7.2 Microphone types M125/SIC and M132/SIC. Type M125/SIC and M132/SIC microphones shall be equipped with 120 inches (extended) of retractable cordage conforming to the general requirements of MIL-DTL-24643/4 for LSMCOS-5 cable, except that it shall be coiled. Type M125/SIC and M132/SIC microphones shall also be provided with a microphone connector plug in accordance with S6502-74117.

3.7.3 Microphone type M131/WIC. The type M131/WIC microphone shall be equipped with a microphone cable of the type specified in 3.7.2 and a type U-77/U microphone connector plug in accordance with MIL-C-10544.

3.7.4 Microphone type M136/SIC. The type M136/SIC microphone shall be equipped with a M124/SIC microphone with a press-to-talk switch housed in a wear-resistant plastic case. The microphone cord shall be encased in a flexible arm consisting of two double-interlocking stainless-steel hoses. The flexible arm shall be attached to a modified 4R size brass enclosure in accordance with MIL-DTL-24558/18. The modified brass enclosure shall include a six-pole terminal block for connections between the microphone and system. The M136/SIC shall include a wall-mounted, stainless-steel microphone holder assembly with a spring-loaded retaining latch for one-handed operation. Figure 2 illustrates the M136/SIC microphone assembly with wall-mounted holder.

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3.8 Press-to-talk switch. The press-to-talk switch shall be a positive-action double pole single-throw switch and shall be capable of closing the control circuit when activated. It shall be mounted in such a manner that the pushbutton is at the top of the microphone.

3.9 Electrical circuit. The electrical circuit of microphone type M132/SIC shall be as shown on figure 3, section a. The electrical circuit of microphone types M124/SIC and M125/SIC shall be as shown on figure 3, section b. The electrical circuit of microphone type M131/WIC shall be as shown on figure 3, section c. The electrical circuit of microphone type M136/SIC shall be as shown on figure 3, section d.

3.10 Wiring diagram. A printed card, tag, or label showing a schematic wiring diagram of the microphone, cord, and plug with all parts named for rapid and easy identification shall be supplied with each microphone.

3.11 Microphone holder assembly, type MT-3400/SIC. Each microphone holder assembly shall be assembled in accordance with figure 1 from the following individual parts:

a. One microphone clamp (figure 1, section a) made of annealed cold rolled nickel-copper alloy with satin finish in accordance with QQ-N-281.

b. One spring (figure 1, section b) made of phosphor bronze material in accordance with ASTM B139/B139M and nickel plated or of nickel-copper alloy in accordance with QQ-N-281.

c. One catch (figure 1, section c) made of annealed cold rolled nickel-copper alloy in accordance with QQ-N-281.

d. Four stainless-steel or nickel-plated brass rivets 0.121 inch in diameter and 5⁄32 inch in length.

3.12 Performance requirements.

3.12.1 Frequency response. The frequency response shall fall within the limits shown by the solid lines on figure 4 when measured in accordance with 4.6.1. The zero-decibel reference level shall be the microphone output voltage measured at a frequency of 1,000 Hertz (Hz).

3.12.2 Microphone output voltage. The output voltage measured in accordance with 4.6.2 shall be a minimum of 1 millivolt root mean square (rms) to a maximum of 6 millivolts rms.

3.12.2.1 Microphone on-axis and off-axis voltage. The on-axis and off-axis voltage shall be measured in accordance with 4.6.2.1 and the measured levels shall fall within the limits shown by the solid lines on figure 4.

3.12.3 Microphone linearity. The difference between the frequency response curves measured in accordance with 4.6.3 shall be 20±1.5 decibels throughout the frequency range from 200 to 6,400 Hz.

3.12.4 Noise cancelling. When tested in accordance with 4.6.4, the noise cancellation characteristic of the microphone shall be such that the average difference curve and the peaks of difference curves A and B, derived from the frequency response characteristics of the microphone to a close and a distant sound source as specified herein, shall fall within an envelope having a slope of no less than 6 decibels per octave falling between 200 and 4,000 Hz. Difference curves A and B may go through the envelope on the lower side. They shall not go through the envelope on the top side. The centerline of the envelope shall intersect the zero axis between 5,000 and 8,000 Hz.

3.12.5 Induced voltage. When the microphone is subjected to the induced voltage tests specified in 4.6.5, the measured induced voltage shall not exceed 20 microvolts.

3.12.6 Microphone endurance. The microphone shall be capable of withstanding the endurance test specified in 4.6.6 without evidence of physical or electrical damage.

3.12.7 Switch endurance. The microphone press-to-talk switch shall be capable of completing 100,000 operations when tested in accordance with 4.6.7 without evidence of mechanical or electrical failure.

3.12.8 Cable assembly strength. There shall be no evidence of physical damage to any of the electrical connections following the cable assembly strength test specified in 4.6.8.

3.12.9 Dielectric strength. The electrical circuits shall be capable of withstanding the dielectric strength test specified in 4.6.9.

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3.12.10 Insulation resistance. Insulation resistance shall be greater than 10 megohms when measured in accordance with 4.6.10.

3.12.11 Extended storage. The microphone shall be capable of withstanding the extended storage test specified in 4.6.11 without evidence of any physical damage.

3.12.12 Salt spray. The microphone shall be capable of withstanding the salt spray test specified in 4.6.12 without evidence of physical damage or corrosion.

3.12.13 Random drop. The microphone shall be capable of withstanding the random drop test specified in 4.6.13 without evidence of physical damage.

3.12.14 Vibration. The microphone shall be capable of withstanding the vibration test specified in 4.6.14 without evidence of physical damage.

3.12.15 Shock. Following the shock test specified in 4.6.15, the microphone shall be capable of normal operation and shall show no evidence of physical damage.

3.12.16 Frequency response after environmental test. The resultant frequency response curve shall fall within the limits defined by the dotted lines shown on figure 4. The output voltage at 1,000 Hz measured in the test performed after the environmental tests specified in 4.6.11 through 4.6.15 shall be used as the zero-decibel reference level in determining conformance to 4.6.16. The output voltage at 1,000 Hz shall not have changed more than 3 decibels from the original value as a result of the series of environmental tests.

3.13 Equipment identification.

3.13.1 Identification of microphones. Each microphone furnished under this specification shall be marked to show the microphone nomenclature and the number of the contract under which the microphone is procured. Microphones shall be marked by a cable band-marker of polyvinylchloride material in accordance with MIL-I-631.

3.13.2 Identification of microphone holder. The microphone holder assembly shall have the designation “Microphone Holder MT-3400/SIC” stamped on the side of the clamp (figure 1, section a).

3.14 Nomenclature. Government assigned nomenclature is required for microphones furnished under this specification. The nomenclature assigned to each type microphone will be essentially that used in this specification except a letter may be added after the number portion of the nomenclature to indicate a specific design. Government nomenclature shall be requested as specified (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on two samples of each microphone type in 1.2 (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.6.1 through 4.6.16. The examination and tests shall be accomplished in the order listed in table I. Previously inspected microphones to which design or material change has occurred shall undergo first article inspection. Deviations from these requirements shall be approved by the concerned bureau or agency (see 6.2).

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TABLE I. First article inspection.

Inspection	Requirement paragraph	Verification paragraph
General	3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.10, 3.11, 3.13, 3.14	4.4 and 4.5
Frequency response	3.12.1	4.6.1
Microphone output voltage	3.12.2	4.6.2
Microphone on-axis and off-axis voltage	3.12.2.1	4.6.2.1
Microphone linearity	3.12.3	4.6.3
Noise cancelling	3.12.4	4.6.4
Induced voltage	3.12.5	4.6.5
Microphone endurance	3.12.6	4.6.6
Switch endurance	3.12.7	4.6.7
Cable assembly strength	3.12.8	4.6.8
Dielectric strength	3.12.9	4.6.9
Insulation resistance	3.12.10	4.6.10
Extended storage	3.12.11	4.6.11
Salt spray	3.12.12	4.6.12
Random drop	3.12.13	4.6.13
Vibration	3.12.14	4.6.14
Shock	3.12.15	4.6.15
Frequency response	3.12.16	4.6.16

4.3 Conformance inspection. Conformance inspection shall be performed on each lot of delivered components. Conformance inspection shall include the examination and tests specified in table II for sampling groups A (see 4.3.2) and B (see 4.3.3).

TABLE II. Conformance inspection.

Inspection	Requirement paragraph	Verification paragraph
Group A		
General	3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.10, 3.11, 3.13, 3.14	4.4 and 4.5
Frequency response	3.12.1	4.6.1
Microphone output voltage	3.12.2	4.6.2
Microphone on-axis and off-axis voltage	3.12.2.1	4.6.2.1
Group B		
Microphone linearity	3.12.3	4.6.3
Noise cancelling	3.12.4	4.6.4

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TABLE II. Conformance inspection – Continued.

Inspection	Requirement paragraph	Verification paragraph
Induced voltage	3.12.5	4.6.5
Dielectric strength	3.12.9	4.6.9
Insulation resistance	3.12.10	4.6.10

4.3.1 Inspection lot. Microphones presented for delivery at the same time shall be considered a lot. The lot may include the entire contract quantity or may be the production of convenient time periods.

4.3.2 Sampling for group A examination and tests. A sample of microphones shall be selected from each inspection lot in accordance with table III and subjected to the group A examination and tests specified in table II. The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A examination or test shall be counted as a defect, and the microphone shall not be offered for delivery. If the number of nonconforming microphones in any sample exceeds the conformance failure limit (i.e., the number of microphones allowed to fail) for that sample (see 6.2), it shall be cause for rejection of the lot represented by that sample.

TABLE III. Sampling for group A examination and tests.

Number of microphones in inspection lot	Number of microphones in sample
7 and under	All
8 to 15	7
16 to 40	10
41 to 110	15
111 to 300	25
301 to 500	35
501 and over	50

4.3.3 Sampling for group B tests. A sample of microphones shall be selected from each inspection lot in accordance with table IV and subjected to the group B tests specified in table II. The results of each test shall be compared with specification requirements. Failure to conform to the specification requirements for any group B test shall be counted as a defect, and the microphone shall not be offered for delivery. If the number of nonconforming microphones in any sample exceeds the Conformance Failure Limit for that sample (see 6.2), it shall be cause for rejection of the lot represented by that sample.

TABLE IV. Sampling for group B tests.

Number of microphones in inspection lot	Number of microphones in sample
3 and under	All
4 to 15	3
16 to 40	5
41 to 110	7
111 to 300	10

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TABLE IV. Sampling for group B tests – Continued.

Number of microphones in inspection lot	Number of microphones in sample
301 to 500	15
501 and over	25

4.4 General examination. The requirements of 3.3 through 3.11, 3.13, and 3.14 shall be verified via a general examination.

4.5 Prohibited materials. Prohibited materials shall be verified as specified (see 3.3.4 and 6.2).

4.6 Test procedures.

4.6.1 Frequency response.

4.6.1.1 Acoustic environment. The acoustic environment in which the frequency response and microphone linearity tests are made shall be free from reflected waves that would affect the measured results by more than one decibel at any test frequency. The maximum permissible acoustic noise level of the environment shall be at least 20 decibels below that produced during a test made in accordance with 4.6.1.4.

4.6.1.2 Sound source. The sound source used for the frequency response and microphone linearity tests shall be a mouth simulator, Bruel & Kjar model 4227, 4227-A, or equal. The mouth simulator shall be capable of producing an rms sound pressure of 80 microbars at a point on-axis and ½ inch from a plane across the front opening of the mouth simulator for any test frequency in the range of 180 to 5,600 Hz with a maximum distortion of 5 percent. The sine-wave signal to be supplied to the mouth simulator amplifier shall be obtained from a continuously variable frequency sine-wave generator. A means shall be provided to keep the sound pressure level constant within ±2 decibels throughout the frequency range from 180 to 5,600 Hz.

4.6.1.3 Sound field calibration. The sound pressure field produced by the mouth simulator shall be calibrated by a laboratory standard microphone. The laboratory standard microphone shall meet the requirements of ANSI/ASA S1.15/Part 1 for laboratory standard microphones. The laboratory standard microphone shall be mounted in front of the mouth simulator so that its axis is collinear with that of the voice. The microphone face shall be 1.2 inches from the front surface of the mouth simulator.

4.6.1.4 Test procedure. The test microphone shall be positioned so that the axis normal to the plane of its opening is collinear with the axis of the mouth simulator. The distance from the microphone face to the front surface of the mouth simulator shall be 1.2 inches. The microphone audio circuit shall be terminated with a 150-ohm non-inductive resistor. The voltage on the voice coil of the mouth simulator shall be adjusted to the value required to produce a sound pressure of 20 microbars at the face of the test microphone with the signal frequency set at 1,000 Hz. The test microphone output voltage shall be the zero-decibel reference level for the microphone frequency response. The audio signal frequency shall be swept through the 180- to 5,600-Hz range and the test microphone output in decibels referenced to the 1,000-Hz level shall be recorded using audio frequency semi-logarithmic scale. The resulting response curve shall be compared with figure 4 to determine compliance with 3.12.1.

4.6.2 Microphone output voltage. The microphone output voltage shall be the voltage measured across the 150-ohm non-inductive resistor with a 1,000-Hz, 20-microbar sound pressure applied to the microphone as specified in 4.6.1.4. Results shall comply with 3.12.2.

4.6.2.1 On-axis and off-axis voltage test. The sound source for this test shall be the mouth simulator specified in 4.6.1.2. The mouth simulator output shall be adjusted to produce a sound pressure of 50 microbars at a point on-axis and 1.2 inches in front of the voice for test frequencies of 400 to 1,000 Hz. The microphone output voltage shall be measured across a 150-ohm non-inductive resistor for each of the following test conditions.

4.6.2.1.1 Close talking on-axis. The microphone shall be mounted in the normal test position as specified in 4.6.1.4.

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4.6.2.1.2 Distant talking off-axis. The microphone shall be positioned in front of and facing the mouth simulator with its principal axis parallel to that of the voice. The position of the microphone shall be such that the center of its face is at a point 3 inches from the center of the front opening of the voice on a line drawn at a 45-degree angle to the principal axis of the mouth simulator.

4.6.2.1.3 Ratio of on-axis to off-axis voltage output. The ratio of output voltage measured with the microphone on-axis to the output voltage measured with the microphone off-axis shall be expressed in decibels. Results shall comply with 3.12.2.1.

4.6.3 Microphone linearity. The frequency response measurement described in 4.6.1.4 shall be repeated for each of the two conditions. For the first test, the reference sound pressure at 1,000 Hz shall be 5 microbars. For the second test, the reference sound pressure at 1,000 Hz shall be 50 microbars. The frequency response at both levels shall be recorded on a single semilogarithmic graph. The results shall comply with 3.12.3.

4.6.4 Noise cancelling test. The noise-cancellation characteristics of the microphone shall be determined by comparing its sensitivity and frequency response characteristics for a close sound source to that for a distant sound source. The results shall comply with 3.12.4.

4.6.4.1 Frequency response and sensitivity to a close sound source for noise-immunity characteristics measurements. The frequency response of the microphone shall be measured in accordance with 4.6.1, and the microphone voltage output shall be measured in accordance with 4.6.2.

4.6.4.2 Frequency response and sensitivity to a distant sound source for noise-cancellation characteristics measurements. The frequency response characteristics and sensitivity to a distant sound source shall be determined by measuring the voltage output of the microphone developed across the 150-ohm non-inductive resistor when the microphone is placed in a calibrated sound field at least 1 meter from a sound source in an anechoic chamber.

4.6.4.2.1 Sound field calibration. The sound pressure field shall be calibrated at a point no less than 1 meter from the sound source using a laboratory standard microphone and free-field calibration. The laboratory standard microphone shall meet the requirements of ANSI/ASA S1.15/Part 1 for laboratory standard microphones. The point where the calibration microphone is placed shall be known as the calibration point.

4.6.4.2.2 Sensitivity to a distant sound source. The sensitivity of the microphone to a distant sound source at the calibration point shall be established by measuring the voltage developed by the microphone across the 150-ohm non-inductive resistor and referenced to voltage from a sound pressure of 1 microbar at 1,000 Hz. The sensitivity measurement shall be made twice, in each case with the cylindrical axis of the microphone element normal to the axis of the sound source. In the first measurement, sensitivity A, the talking port of the microphone shall point away from the sound source. In the second measurement, sensitivity B, the talking port of the microphone shall point toward the sound source. In each measurement, the microphone shall be adjusted about the cylindrical axis of the microphone element to obtain maximum outputs.

4.6.4.2.3 Frequency response to a distant sound source. The frequency response of the microphone at the calibration point shall be established by measuring the voltage developed by the microphone across the 150-ohm non-inductive resistor as a continuous function of frequency from 200 to 6,000 Hz. It shall be referenced to voltage from a sound field of constant 1-microbar sound pressure as determined with the standard condenser microphone. Two frequency response curves shall be made, in each case with the cylindrical axis of the microphone element normal to the axis of the sound source. The frequency response obtained with the talking port of the microphone pointing away from the sound source shall be known as distant response A and that obtained with the talking port pointing toward the sound source shall be known as distant response B.

4.6.4.3 Formation of noise cancellation characteristics.

4.6.4.3.1 Plotting of data. The close frequency response characteristics obtained herein shall be plotted on a semi-logarithmic graph with the length of a 10 to 1 frequency interval along the abscissa equal to the length of 30 decibels along the ordinate. The distant response A shall be plotted on the same coordinates with the point on the curve at 1,000 Hz located below the point on the close response curve at 1,000 Hz by the number of decibels equivalent to the ratio of the close sensitivity in volts at 1,000 Hz to the distant sensitivity in volts at 1,000 Hz.

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$$\text{Number of dB} = 20 \ \text{LOG}_{10} \left[\frac{\text{Close sensitivity at 1,000 Hz}}{\text{Distant sensitivity at 1,000 Hz}} \right]$$

The same procedure shall be followed with distant response B and distant sensitivity B. A fourth curve shall be obtained by averaging the ordinates of distant response A and B at each point between 200 and 6,000 Hz and called the average distant response curve.

4.6.4.3.2 Construction of difference curves. The three curves of distant response A, distant response B, and the average distant response shall be individually subtracted from the close response curve and the differences plotted. These shall be called, respectively, difference curve A, difference curve B, and average difference curve.

4.6.5 Electromagnetically induced voltage. The microphone that is being tested shall be complete with a microphone cable and connector plug. A 150-ohm non-inductive resistor shall be connected to the microphone audio circuit at the connector end of the cable. The press-to-talk switch shall be closed and a 0.5-ampere rms 60-Hz sinusoidal current shall be made to flow through the control circuit leads. The induced voltage shall be measured across the 150-ohm non-inductive resistor. Results shall comply with 3.12.5.

4.6.6 Microphone endurance test. A warble frequency signal of 800 to 1,250 Hz at a voltage of 25 millivolts rms shall be applied across the microphone voice coil for an uninterrupted period of 8 hours with the microphone in an ambient room temperature of 72±5 °F. Results shall comply with 3.12.6.

4.6.7 Switch endurance test. The microphone control circuits shall be connected to a 115-volt 60-Hz supply through suitable resistance and inductance to cause a current 0.5 amperes at 50 percent power factor to flow in the control circuit when the press-to-talk switch is closed. The press-to-talk switch shall be operated 100,000 times to determine compliance with 3.12.7.

4.6.8 Cable assembly strength test. A microphone holder assembly of the type specified in 3.11 shall be mounted on a firm surface. The microphone being tested shall be complete with a cable and plug and shall be locked into the microphone holder assembly. A force of 20 pounds shall be applied to the connector plug in the direction facing away from the microphone case and along a line that is coincident with the axis of the portion of cable that is clamped in the microphone case. Results shall comply with 3.12.8.

4.6.9 Dielectric strength test. Dielectric strength between electrical circuits and between electrical circuits and external metal parts of the microphone shall be tested in accordance with MIL-STD-202-301. Results shall comply with 3.12.9.

4.6.10 Insulation resistance. The insulation resistance of the electrical circuits of the microphone shall be measured at a room temperature of 72±5 °F and at a relative humidity of approximately 50 percent. The insulation resistance shall be measured using a voltage of 50 or more volts direct current to determine compliance with 3.12.10.

4.6.11 Extended storage test. The microphone shall be capable of extended storage (2 years or longer) in any ambient temperature between the limits of -40 and 167 °F. MIL-STD-810, Method 501 (High Temperature), Procedure I (Storage), with 167 °F as the maximum temperature shall be used for high temperature cycles to test the upper storage limit. Method 502 (Low Temperature), Procedure I (Storage), with -40 °F as the test temperature, shall be used for testing the lower storage limit. The audible signals shall be subjected to the frequency response test specified in 4.6.1 and the microphone output voltage test specified in 4.6.2 subsequent to stabilization at room temperature. Results shall comply with 3.12.11.

4.6.12 Salt spray test. The microphone shall be subjected to a salt spray test performed in accordance with MIL-STD-202-101. The salt solution concentration shall be 20 percent. The length of the test shall be 96 hours. Results shall comply with 3.12.12.

4.6.13 Random drop test. The microphone shall be subjected to 10 random drops from a 6-foot height onto a concrete surface. Results shall comply with 3.12.13.

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4.6.14 Vibration test. The microphone shall be placed in a microphone holder assembly of the type specified in 3.11, which is mounted on a simulated bulkhead. The microphone shall then be subjected to type I (environmental vibration) testing in accordance with MIL-STD-167-1 except that the microphone circuits shall not be energized during the test and the variable frequency test shall be omitted. Results shall comply with 3.12.14.

4.6.15 Shock test. The microphone shall be placed in a microphone holder assembly of the type specified in 3.11, which is mounted on a simulated bulkhead. The microphone shall then be subjected to the type A, high impact shock test in accordance with MIL-DTL-901 except that the microphone circuits shall not be energized during the test. Results shall comply with 3.12.15. Microphones that have been subjected to the shock test shall not be installed aboard ship.

4.6.16 Frequency response after environmental tests. At the conclusion of the series of environmental tests specified in 4.6.11 through 4.6.15, the frequency response test specified in 4.6.1.4 shall be repeated. Results shall comply with 3.12.16.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The microphones covered by this specification are intended for use with intercommunication and announcing equipment aboard naval ships. Existing stock of microphones manufactured to MIL-M-23813A is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of items required.

c. Type required (see 1.2).

d. Exceptions or deviations to first article inspection (see 3.1).

e. Exception if a microphone holder assembly is not required with each microphone (see 3.2).

f. Other aluminum material approved for construction (see 3.3.1.1).

g. Requirement for prohibited materials verification (see 3.3.4 and 4.5).

h. Requirement to request Government nomenclature (see 3.14).

i. Conformance failure limit (see 4.3.2 and 4.3.3) (typical values: 0 for lot size 1-110; 1 for lot size 111-500; 2 for lot size >500).

j. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Press-to-talk

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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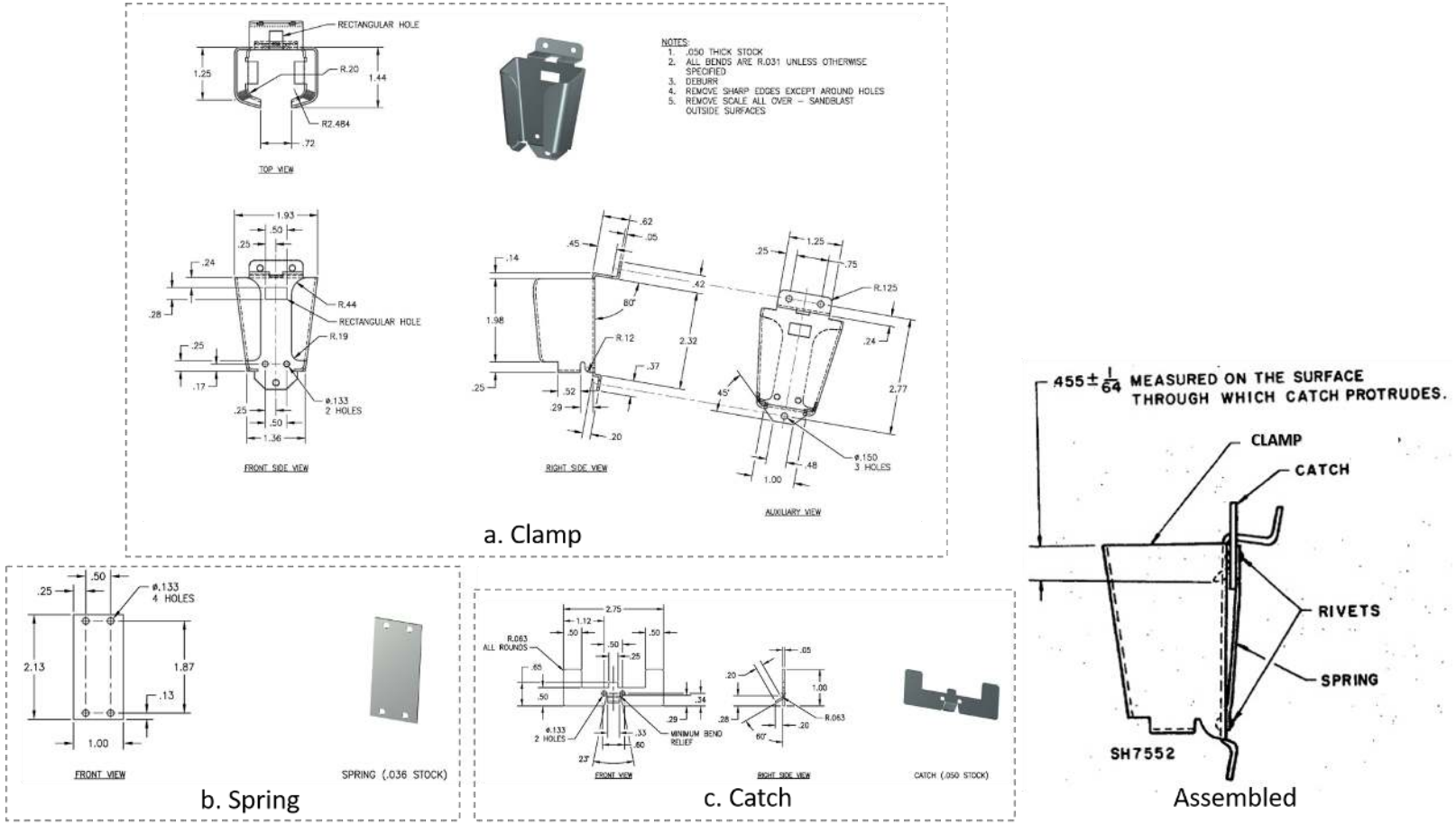


FIGURE 1. MT-3400/SIC microphone holder assembly.

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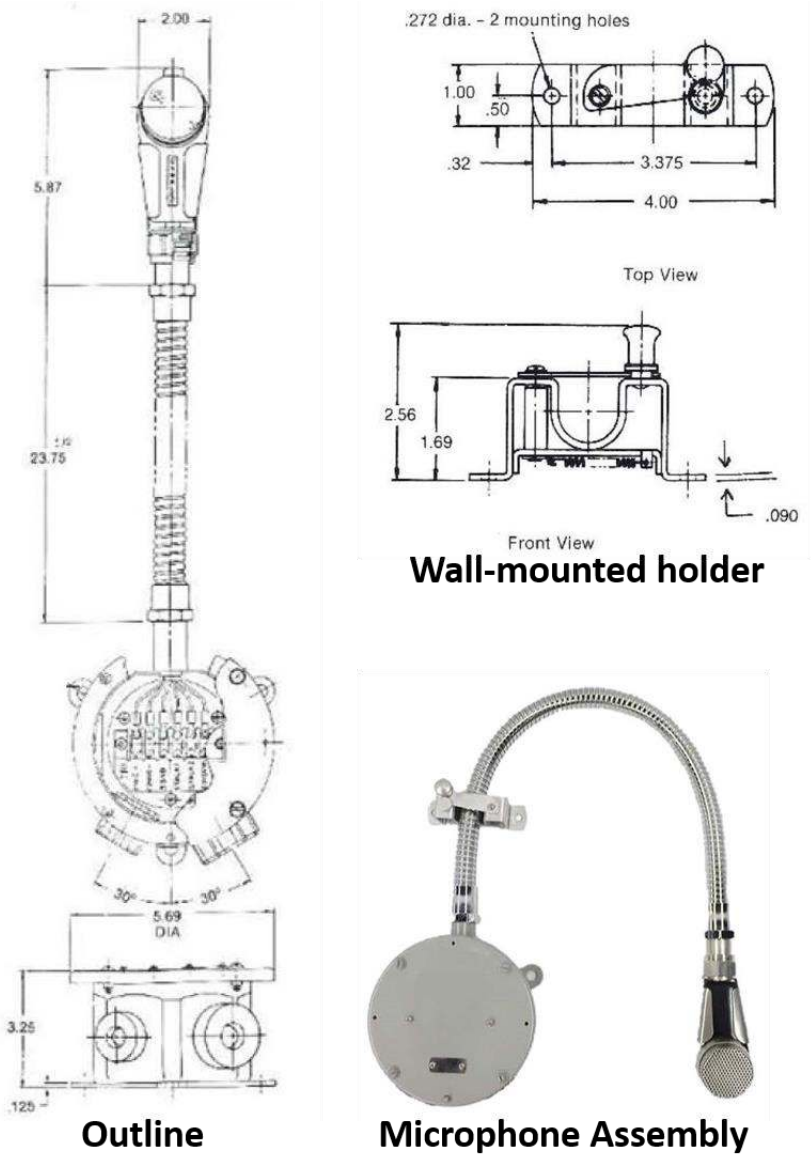


FIGURE 2. M136/SIC microphone assembly.

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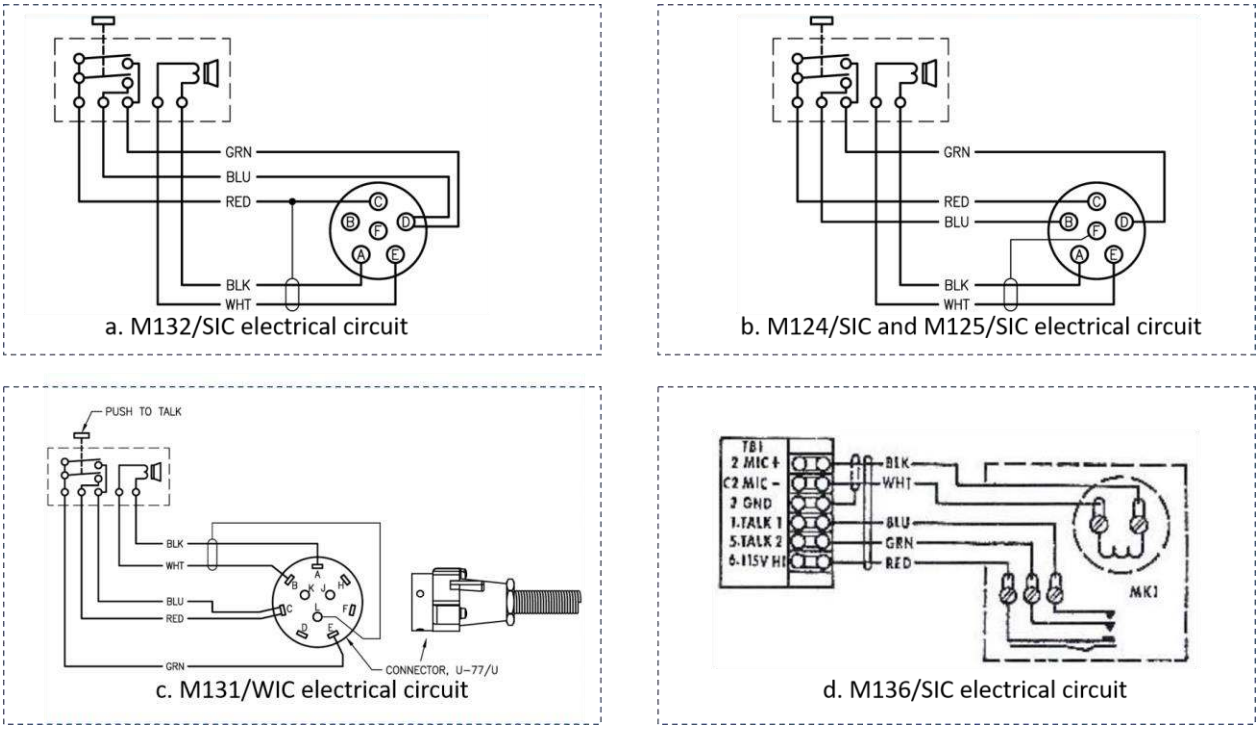


FIGURE 3. Microphone electrical circuits.

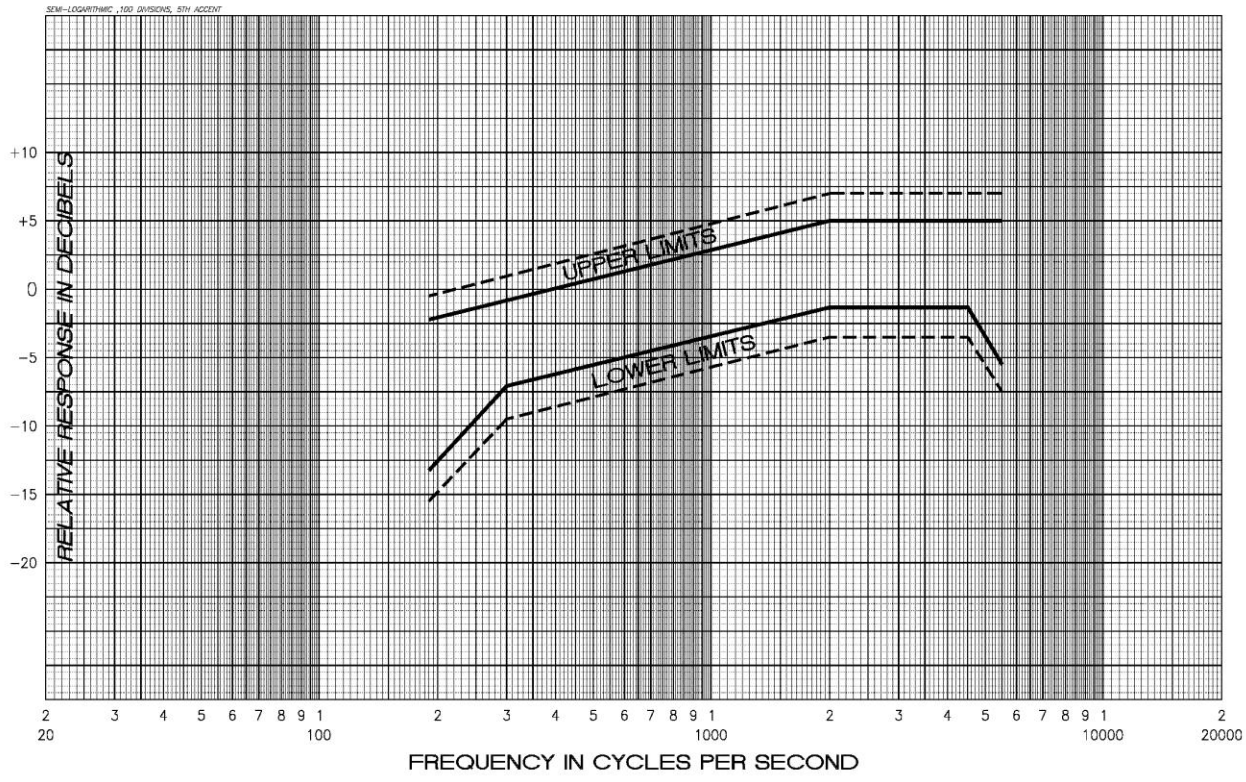


FIGURE 4. Frequency response limits for microphones.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH

Preparing activity:
Navy – SH
(Project 5830-2021-001)

Review activity:
Navy – MC

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